COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 / Data Structures and Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	08/07/2026	**DATE SUBMITTED:**	7/9/2026
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ACTIVITY NO. 1.1 **Algorithm, Flowchart, Pseudocode**

1. PROCEDURE OUTPUT

1. Compute the total and average of 3 quizzes then display the total and average.

Algorithm

1. Start
2. Enter the values of the variables of Score 1, Score 2, Score 3
3. Sum them and save the result in variable Total
4. Divide the value of Total to 3 and save the output to Average
5. Display the Average
6. End

Pseudocode

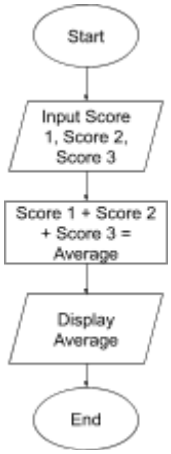
Start
Input Score 1, Score 2, Score 3

Score 1 + Score 2 + Score 3/3 = Average

Print Average

End

Flow Chart



Code

```
using namespace std;
#include <iostream>

int main () {

  // To store values of quiz scores
  int ScoreOne;
  int ScoreTwo;
  int ScoreThree;
  float Average;

  cout <<"Welcome please choose one of the following options: " <<endl;

  // The section that takes the values of the quiz scores and stores them
  cout <<"Please enter the score of quiz 1: ";
  cin >> ScoreOne;
  cout <<"Please enter the score of quiz 2: ";
  cin >> ScoreTwo;
  cout <<"Please enter the score of quiz 3: ";
  cin >> ScoreThree;

  //Calculates the average of the quiz scores
  Average = (ScoreOne + ScoreTwo + ScoreThree)/3;

  //Displays the quiz scores
  cout << Average;

  return 0;
}
```

2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes change. Display the transactions made.

Algorithm

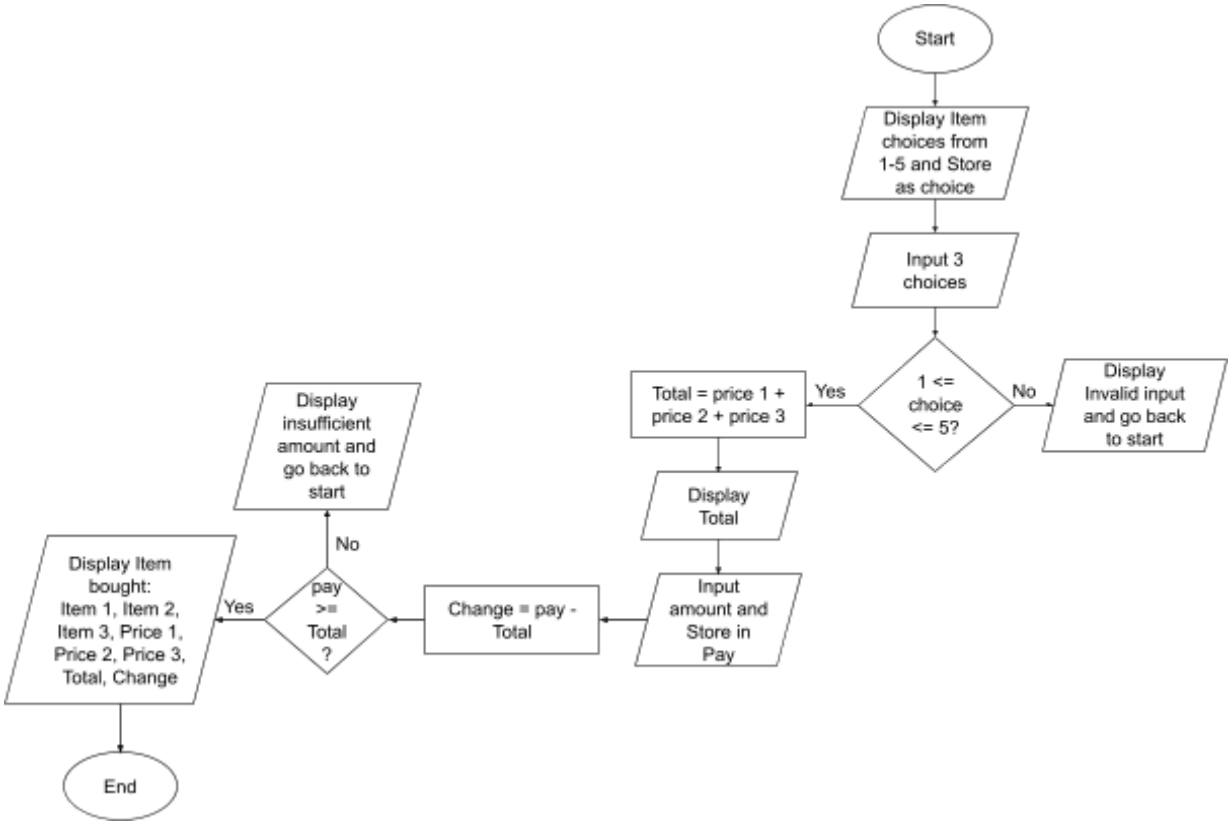
1. Start
2. Enter three of the items you want to buy from the list and store them in the variable Choice1, Choice2, Choice3.
3. Each items have their own price that is stored in variable Price1, Price2, Price3
4. If input given is are contained inside the options proceed, if not go back to start
5. Sum the chosen items in their respected prices then save result to Total
6. Display Total to be paid

7. Input payment amount and save as AmountPaid
8. Subtract Total from AmountPaid and save as Change
9. Display Total, AmountPaid, and Change
10. End

Pseudocode

Start
Display options
Input Item1, Item2, Item3
Are the inputs only from 1-5? If yes then proceed if not display error and go back to start
Total = price1 + price2 + price3
Display total to be paid
Input payment amount and save as Pay
Is Input paid amount (pay >= Total) If yes proceed if not Display error and go back to start
Display Item1, Item2, Item3 and Price1, Price2, Price3 and finally Total and Change
End

Flow Chart



Code

```cpp
#include <iostream>
using namespace std;

int main() {

    //Stores the values
    string Item1;
    string Item2;
    string Item3;
    int price1;
    int price2;
    int price3;
    int choice1;
    int choice2;
    int choice3;
    int Total;
    int pay;

    cout << "Welcome! Please choose 3 items of your choice to purchase." << endl << endl;
    cout << "1.) Apple 30?" << endl << "2.) Orange 24?" << endl << "3.) Pineapple 68?" << endl << "4.) Watermelon 86?" << endl << "5.) Mango 50?" << endl << endl;

    //Gets the value for choices 1, 2, and 3
    cout << "Please input from 1-5: ";
    cin >> choice1;

    switch (choice1) {
        case 1: price1 = 30;
        Item1 = "Apple";
        break;
        case 2: price1 = 24;
        Item1 = "Orange";
        break;
        case 3: price1 = 68;
        Item1 = "Pineapple";
        break;
```

```
    break;
    case 4: price1 = 86;
    Item1 = "Watermelon";
    break;
    case 5: price1 = 50;
    Item1 = "Mango";
    break;

    // Is the default outcome when users inputs a value other then the numbers 1-5
    default: cout << "Invalid Input please try again and choose from 1-5";
    return 0;
}

cout << "Please input from 1-5: ";
cin >> choice2;
switch (choice2) {
    case 1: price2 = 30;
    Item2 = "Apple";
    break;
    case 2: price2 = 24;
    Item2 = "Orange";
    break;
    case 3: price2 = 68;
    Item2 = "Pineapple";
    break;
    case 4: price2 = 86;
    Item2 = "Watermelon";
    break;
    case 5: price2 = 50;
    Item2 = "Mango";
    break;

    default: cout << "Invalid Input please try again and choose from 1-5";
    return 0;
```

```
}

cout << "Please input from 1-5: ";
cin >> choice3;
cout << " " << endl << endl;
switch (choice3) {
    case 1: price3 = 30;
    Item3 = "Apple";
    break;
    case 2: price3 = 24;
    Item3 = "Orange";
    break;
    case 3: price3 = 68;
    Item3 = "Pineapple";
    break;
    case 4: price3 = 86;
    Item3 = "Watermelon";
    break;
    case 5: price3 = 50;
    Item3 = "Mango";
    break;

    default: cout << "Invalid Input please try again and choose from 1-5";
    return 0;
}
//Gets the total sum of all prices
Total = price1 + price2 + price3;


cout << "Total to be paid: " << Total << " " << "PHP" << endl;
cout << "Items:" << endl << Item1 << " " <<  price1 << " PHP" << endl << Item2 << " " << price2 << " PHP" << endl << Item3 << " " << price3 << " PHP" << endl;
cout << "Input proper amount to be paid: " << endl;
cin >> pay;


   //Checks if amount input is sufficient
   if (pay >= Total) {
       int Change = pay - Total;

       //Displays all Items bought their prices and the total amount to be paid, amount of payment given, and change
       cout << endl << "Item bought: " << endl << Item1 << " " <<  price1 << " PHP" << endl << Item2 << " " << price2 << " PHP" << endl << Item3 << " " << price3 << " PHP" << endl << "Total: " << Total <

   }
   else {
       cout << "Input amount insufficient try again";
   }

   return 0;
}
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Recalling functions**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Used to give different examples on to recall the properly use of different functions**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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